

IOWA STATE UNIVERSITY

LAB 2 - BEAM BY CHART REPORT

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Aer E 4260 Design of Aerospace Structures

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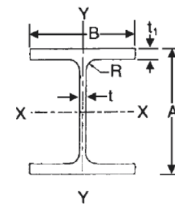
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INTRODUCTION

In Lab 2, we were tasked with designing a beam capable of picking up a 100 lb satellite in space. The beam was modeled as a cantilever with one end fixed and the satellite applied as a point load at the free end. The key design constraint was limiting the maximum deflection to no more than 1/25th of an inch. We performed calculations using two different materials—aluminum and steel—each with specified material properties. The design required us to select the best beam for each material from a supplier's catalog, research the prices, and determine the most cost-effective and structurally sound option. In addition to calculating the factors of safety, we considered whether the beam's weight impacted the design.

BEAM GEOMETRY AND MATERIALS PROPERTIES



**TABLE 13.4 Aluminum Association Standard I-Beams—
Dimensions, Areas, Weights and Section Properties ④**

Size		Area in. ²	Weight lb./ft.	Flange Thickness t ₁ in.	Web Thickness t in.	Fillet Radius R in.	Section Properties ③					
Depth A in.	Width B in.						Axis X-X			Axis Y-Y		
							I in. ⁴	S in. ³	r in.	I in. ⁴	S in. ³	r in.
3.00	2.50	1.392	1.637	0.20	0.13	0.25	2.24	1.49	1.27	0.52	0.42	0.61
3.00	2.50	1.726	2.030	0.26	0.15	0.25	2.71	1.81	1.25	0.68	0.54	0.63
4.00	3.00	1.965	2.311	0.23	0.15	0.25	5.62	2.81	1.69	1.04	0.69	0.73
4.00	3.00	2.375	2.793	0.29	0.17	0.25	6.71	3.36	1.68	1.31	0.87	0.74
5.00	3.50	3.146	3.700	0.32	0.19	0.30	13.94	5.58	2.11	2.29	1.31	0.85
6.00	4.00	3.427	4.030	0.29	0.19	0.30	21.99	7.33	2.53	3.10	1.55	0.95
6.00	4.00	3.990	4.692	0.35	0.21	0.30	25.50	8.50	2.53	3.74	1.87	0.97
7.00	4.50	4.932	5.800	0.38	0.23	0.30	42.89	12.25	2.95	5.78	2.57	1.08
8.00	5.00	5.256	6.181	0.35	0.23	0.30	59.69	14.92	3.37	7.30	2.92	1.18
8.00	5.00	5.972	7.023	0.41	0.25	0.30	67.78	16.94	3.37	8.55	3.42	1.20
9.00	5.50	7.110	8.361	0.44	0.27	0.30	102.02	22.67	3.79	12.22	4.44	1.31
10.00	6.00	7.352	8.646	0.41	0.25	0.40	132.09	26.42	4.24	14.78	4.93	1.42
10.00	6.00	8.747	10.286	0.50	0.29	0.40	155.79	31.16	4.22	18.03	6.01	1.44
12.00	7.00	9.925	11.672	0.47	0.29	0.40	255.57	42.60	5.07	26.90	7.69	1.65
12.00	7.00	12.153	14.292	0.62	0.31	0.40	317.33	52.89	5.11	35.48	10.14	1.71

① Areas listed are based on nominal dimensions.

② Weights per foot are based on nominal dimensions and a density of 0.098 pound per cubic inch which is the density of alloy 6061.

③ I = moment of inertia; S = section modulus; r = radius of gyration.

④ Users are encouraged to ascertain current availability of particular structural shapes through inquiries to their suppliers.

Table 2.1: Frontal view of the beam and table of available aluminum beams and their properties.

Steel Beam

Beam height h, Beam width b, thickness of web tw and thickness of flange tf.

	Item#		h	tw	b	tf	Area	Ixx	Iyy
		lb/ft	in	in	in	in	in ²	in ⁴	in ⁴
1	S24x121	3615.59	24.5	0.8	8.05	1.09	35.6	3634.7051	95.8136
2	S24x106	3336.08	24.5	0.62	7.87	1.09	31.2	3354.7625	89.0388
3	S24x100	2674.34	24	0.745	7.245	0.87	29.3	2683.0889	55.9691
4	S24x90	2506.02	24	0.625	7.125	0.87	26.5	2514.6237	52.9355
5	S24x80	2330.69	24	0.5	7	0.87	23.5	2339.1390	49.9850
6	S20x96	1923.47	20.3	0.8	7.2	0.92	28.2	1933.7503	58.0975
7	S20x86	1799.32	20.3	0.66	7.06	0.92	25.3	1809.3970	54.4437
8	S20x75	1439.08	20	0.635	6.385	0.795	22	1444.9647	34.9172
9	S20x66	1331.74	20	0.505	6.255	0.795	19.4	1337.4974	32.6410
10	S18x70	1045.64	18	0.711	6.251	0.691	20.6	1049.4203	28.6695
11	S18x54.7	896.14	18	0.461	6.001	0.691	16.1	899.7699	25.0354
12	S15x50	549.57	15	0.55	5.64	0.622	14.7	552.0609	18.8064
13	S15x42.9	500.75	15	0.411	5.501	0.622	12.6	503.1738	17.3437
14	S12x50	359.06	12	0.687	5.477	0.659	14.7	361.9356	18.3695
15	S12x40.8	315.98	12	0.462	5.252	0.659	12	318.7310	16.0100
16	S12x35	260.66	12	0.428	5.078	0.544	10.3	262.1621	11.9505
17	S12x31.8	246.37	12	0.35	5	0.544	9.4	247.8499	11.3762
18	S10x35	170.97	10	0.594	4.944	0.491	10.3	172.0457	10.0640
19	S10x25.4	140.44	10	0.311	4.661	0.491	7.46	141.4477	8.3115
20	S8x23	75.73	8	0.441	4.171	0.426	6.77	76.3200	5.2092
21	S8x18.4	66.16	8	0.271	4.001	0.426	5.41	66.7229	4.5607
22	S7x20	49.97	7	0.45	3.86	0.392	5.88	50.3990	3.8106
23	S6x17.25	31.43	6	0.465	3.565	0.359	5.07	31.7369	2.7612
24	S6x12.5	25.73	6	0.232	3.332	0.359	3.67	26.0157	2.2196
25	S5x14.75	18.55	5	0.494	3.284	0.326	4.34	18.7557	1.9745
26	S5x10	14.49	5	0.214	3.004	0.326	2.94	14.6786	1.4770
27	S4x9.5	8.30	4	0.326	2.796	0.293	2.79	8.4332	1.0790
28	S4x7.7	7.28	4	0.193	2.663	0.293	2.26	7.4054	0.9246
29	S3x7.5	3.73	3	0.349	2.509	0.26	2.21	3.8090	0.6950
30	S3x5.7	3.12	3	0.17	2.33	0.26	1.67	3.1905	0.5494

Table 2.2: Table of available steel beams and their properties.

METHODOLOGY

For simplification purposes, it was assumed that the satellite point load would only act along the y direction. Thus, the calculations were only made using I_{xx} .

To address the primary design constraint (maximum deflection of 1/25 inches), we evaluate equation (3.1) at the point load location (free end of the beam). It relates deflection to the load, length, Young's modulus, and moment of inertia about the center x-axis:

$$\delta = \frac{PL^3}{3EI} \quad (3.1)$$

Rearranging (3.1), we derive an expression for the minimum required moment of inertia about the center x-axis:

$$I_{\min} = \frac{PL^3}{3E\delta_{\max}} \quad (3.2)$$

From equation (3.2), we can calculate the minimum required moment of inertia based on the maximum allowable deflection, known length, Young's modulus, and applied point load.

3.1 Aluminum Beam

Starting our design with the aluminum beam, we were given its Young's modulus $E = 10$ msi. By plugging in the values and converting units, we calculate the minimum required moment of inertia about the x-axis:

$$I_{\min, Al} = \frac{(0.1 \text{ klb}) \times (72 \text{ in})^4}{3 \times (10^4 \text{ ksi}) \times (1/25 \text{ in})} = 31.104 \text{ in}^4 \quad (3.3)$$

From Table 2.1, the beam with the smallest moment of inertia greater than $I_{\min}$ is the one with depth 7 in and width 4.5 in. This beam had a moment of inertia $I = 42.89 \text{ in}^4$ and a weight $w = 5.800 \text{ lb/ft}$. We then calculated the deflection due to its weight, the deflection due to the satellite load, and checked if their sum is less than $1/25 = 0.04 \text{ in}$:

$$\delta_{\text{body}} = \frac{5.8 \text{ lb/ft} \times (72 \text{ in})^4}{8 \times (10^7 \text{ psi}) \times (42.89 \text{ in}^4)} = 0.00379 \text{ in} \quad (3.4)$$

$$\delta = \frac{(0.1 \text{ klb}) \times (72 \text{ in})^3}{3 \times (10^4 \text{ ksi}) \times (42.89 \text{ in}^4)} = 0.0290 \text{ in} \quad (3.5)$$

$$\delta_{\text{body}} + \delta = 0.03279 \text{ in} < \frac{1}{25} \text{ in} \quad (3.6)$$

Thus, the total displacement due to both the beam's weight and the satellite load does not exceed the maximum allowable deflection.

With the known dimensions of the chosen I-beam (depth $A = 7 \text{ in}$, width $B = 4.5 \text{ in}$, flange thickness $t_f = 0.38 \text{ in}$, web thickness $t = 0.23 \text{ in}$), we can calculate maximum axial stress and maximum shear stress:

$$M = P \times L = 100 \text{ lb} \times 72 \text{ in} = 7200 \text{ lb} \cdot \text{in} \quad (3.7)$$

Maximum axial stress:

$$\sigma_{\text{max}} = \frac{M \times c}{I} = \frac{(7200 \text{ lb} \cdot \text{in}) \times (7 \text{ in}/2)}{42.89 \text{ in}^4} = 0.588 \text{ ksi} \quad (3.8)$$

Moment of area:

$$Q = B \times t_f \times \frac{A}{2} + t \times \frac{A^2}{8} = (4.5 \text{ in}) \times (0.38 \text{ in}) \times \frac{7 \text{ in}}{2} + (0.23 \text{ in}) \times \frac{(7 \text{ in})^2}{8} = 7.394 \text{ in}^3 \quad (3.9)$$

Shear force:

$$V = P = 100 \text{ lb} \quad (3.10)$$

Maximum shear stress:

$$\tau_{\text{max}} = \frac{V \times Q}{I \times t} = \frac{100 \text{ lb} \times 7.394 \text{ in}^3}{42.89 \text{ in}^4 \times 0.23 \text{ in}} = 0.075 \text{ ksi} \quad (3.11)$$

The factors of safety are calculated as follows:

$$\text{FS}_{\sigma_y} = \frac{\sigma_y}{\sigma_{\text{max}}} = \frac{40 \text{ ksi}}{0.588 \text{ ksi}} = 68.1 \quad (3.12)$$

$$\text{FS}_{\tau_u} = \frac{\tau_u}{\tau_{\text{max}}} = \frac{15 \text{ ksi}}{0.075 \text{ ksi}} = 200 \quad (3.13)$$

$$\text{FS}_{\delta} = \frac{\delta_{\text{max}}}{\delta_{\text{body}} + \delta} = \frac{0.04 \text{ in}}{0.03279 \text{ in}} = 1.22 \quad (3.14)$$

3.2 Steel Beam

We repeat the process for the steel beam with $E = 30 \text{ msi}$:

$$I_{\text{min, Steel}} = \frac{(0.1 \text{ klb}) \times (72 \text{ in})^4}{3 \times (3 \times 10^4 \text{ ksi}) \times (1/25 \text{ in})} = 10.368 \text{ in}^4 \quad (3.15)$$

From Table 2.2, the steel beam with the smallest moment of inertia greater than I_{min} is S5x10. This beam has a moment of inertia $I = 14.6786 \text{ in}^4$ and a weight $w = 14.49 \text{ lb/ft}$.

$$\delta_{\text{body}} = \frac{14.49 \text{ lb/ft} \times (72 \text{ in})^4}{8 \times (3 \times 10^7 \text{ psi}) \times 14.6786 \text{ in}^4} = 0.00921 \text{ in} \quad (3.16)$$

$$\delta = \frac{(0.1 \text{ klb}) \times (72 \text{ in})^3}{3 \times (3 \times 10^4 \text{ ksi}) \times 14.6786 \text{ in}^4} = 0.0283 \text{ in} \quad (3.17)$$

$$\delta_{\text{body}} + \delta = 0.0375 \text{ in} < \frac{1}{25} \text{ in} \quad (3.18)$$

Calculating maximum axial stress and maximum shear stress for the steel beam:

$$\sigma_{\text{max}} = \frac{M \times c}{I} = \frac{(7200 \text{ lb} \cdot \text{in}) \times (5 \text{ in}/2)}{14.6786 \text{ in}^4} = 1.226 \text{ ksi} \quad (3.19)$$

First moment of area:

$$Q = B \times t_f \times \frac{A}{2} + t \times \frac{A^2}{8} = (4 \text{ in}) \times (0.273 \text{ in}) \times \frac{5 \text{ in}}{2} + (0.214 \text{ in}) \times \frac{(5 \text{ in})^2}{8} = 3.117 \text{ in}^3 \quad (3.20)$$

Maximum shear stress:

$$\tau_{\text{max}} = \frac{V \times Q}{I \times t} = \frac{100 \text{ lb} \times 3.117 \text{ in}^3}{14.6786 \text{ in}^4 \times 0.214 \text{ in}} = 0.099 \text{ ksi} \quad (3.21)$$

The factors of safety for the steel beam are:

$$\text{FS}_{\sigma_y} = \frac{\sigma_y}{\sigma_{\text{max}}} = \frac{62 \text{ ksi}}{1.226 \text{ ksi}} = 50.6 \quad (3.22)$$

$$\text{FS}_{\tau_u} = \frac{\tau_u}{\tau_{\text{max}}} = \frac{26 \text{ ksi}}{0.099 \text{ ksi}} = 262 \quad (3.23)$$

$$\text{FS}_{\delta} = \frac{\delta_{\text{max}}}{\delta_{\text{body}} + \delta} = \frac{0.04 \text{ in}}{0.0375 \text{ in}} = 1.07 \quad (3.24)$$

CONCLUSION

In this analysis, we designed a 6-ft long beam to lift a 100 lb satellite in space, comparing the use of aluminum and steel I-beams based on deflection constraints, weight, and cost. After conducting calculations for both materials, we found that both aluminum and steel beams satisfied the design requirements, including deflection limits, safety factors, and load-carrying capacity. However, a cost comparison was necessary to determine the optimal material for a space-bound application.

From online research, we determined the cost of a 6-ft aluminum beam to be \$355.54, while the same size steel beam cost \$207.50. On the surface, this makes steel appear to be the cheaper option. However, the weight of the aluminum beam is 34.8 lb, compared to the steel beam's 86.94 lb. Given the approximate cost to launch 1 lb of material to space using the Falcon 9 rocket is around \$1200, the total launch cost for the aluminum beam would be approximately \$41,760, while for the steel beam, it would be \$104,328. This results in a difference of roughly \$63,000 more to launch the steel beam.

Considering that both beams meet the deflection and strength requirements, and factoring in the significant difference in launch costs, it is far more cost-effective to use the aluminum beam for this application. Its lower weight makes it the superior choice, despite its higher upfront material cost. Therefore, the aluminum beam is the recommended solution for minimizing both structural and financial burdens in space applications.

| A

APPENDIX A

A.1 Rough Work

AerE 426/526
Lab Exercise -2

Due one week after the lab.

You have to design a 6-ft long beam to pick a satellite in space. The satellite weighing 100 lb. To keep the distortion to a minimum, the beam should not move more than 1/25 of an inch when the satellite is picked. Assume the supported end is fixed and the satellite end is free.

$$= 0.04 \text{ in}$$

- Start with an aluminum beam, but you can choose only the cross-section based on the availability from your supplier. Their catalog page is attached. Use $E=10 \text{ msi}$, $\sigma_y=40 \text{ ksi}$, and $\tau_u=15 \text{ ksi}$.
- Also, you can build it with steel, catalog page on back included Use $E=30 \text{ msi}$, $\sigma_y=62 \text{ ksi}$, and $\tau_u=26 \text{ ksi}$.
- Density of Al. 0.1 lb/in^3 and Steel 0.284 lb/in^3 .
- Design the best aluminum and steel beams and calculate their weight and costs. Check if Metal Depot has the size you want. You may get the price from there, or any other supplier.
- Report all factors of safety of your final design.
- Does the weight of the beam make any difference to your design? (δ_{body} is the displacement due to weight w per unit length) Equations: $\delta_{\text{max}} = \frac{PL^3}{3EI}$; $\sigma_{\text{max}} = \frac{Mc}{I}$; and $\tau_{\text{max}} = \frac{VQ}{It}$; $\delta_{\text{body}} = \frac{wL^4}{8EI}$; $Q = B \cdot t_1 \cdot (A/2) + (A/2) \cdot t \cdot (A/4)$

Aluminum Beams

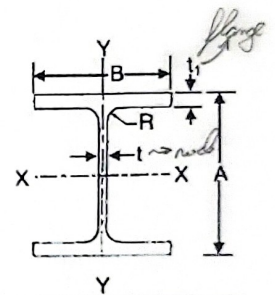
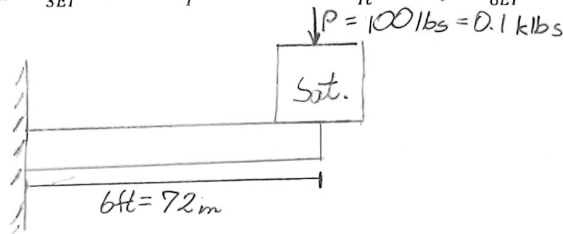


TABLE 13.4 Aluminum Association Standard I-Beams—
Dimensions, Areas, Weights and Section Properties ④

Size		Area ① in. ²	Weight ② lb./ft.	Flange Thickness t _f in.	Web Thickness t _w in.	Fillet Radius R in.	Section Properties ③					
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6.00	4.00	3.990	4.692	0.35	0.21	0.30	25.50	8.50	2.53	3.74	1.87	0.97
7.00	4.50	4.932	5.800	0.38	0.23	0.30	42.89	12.25	2.95	5.78	2.57	1.08
8.00	5.00	5.256	6.181	0.35	0.23	0.30	59.69	14.92	3.37	7.30	2.92	1.18
8.00	5.00	5.972	7.023	0.41	0.25	0.30	67.78	16.94	3.37	8.55	3.42	1.20
9.00	5.50	7.110	8.361	0.44	0.27	0.30	102.02	22.67	3.79	12.22	4.44	1.31
10.00	6.00	7.352	8.646	0.41	0.25	0.40	132.09	26.42	4.24	14.78	4.93	1.42
10.00	6.00	8.747	10.286	0.50	0.29	0.40	155.79	31.16	4.22	18.03	6.01	1.44
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① Areas listed are based on nominal dimensions.

② Weights per foot are based on nominal dimensions and a density of 0.098 pound per cubic inch which is the density of alloy 6061.

③ I = moment of inertia; S = section modulus; r = radius of gyration.

④ Users are encouraged to ascertain current availability of particular structural shapes through inquiries to their suppliers.

$$\delta_{\text{max}} = \frac{PL^3}{3EI} \Rightarrow I_{\text{min}} = \frac{PL^3}{3E\delta_{\text{max}}}$$

$$I_{\text{min}, Al} = \frac{(0.1 \text{ kips})(72 \text{ in})^3}{3(10^4 \text{ ksi})(1/25 \text{ in})} = 31.104 \text{ in}^4$$

$$\delta_{\text{body}} = \frac{(5.8/12 \text{ lb/in})(72 \text{ in})^4}{8(10^7 \text{ psi})(42.89 \text{ in}^4)} = 0.00379 \text{ in}$$

$$\delta_{\text{max}} = \frac{(0.1 \text{ kips})(72 \text{ in})^3}{3(10^4 \text{ ksi})(42.89 \text{ in}^4)} = 0.0290 \text{ in}$$

$$\delta_{\text{max}} + \delta_{\text{body}} = 0.03279 \text{ in} < 1/25 \text{ in}$$

\$355.54 for 6ft beam

$w = 37.8 \text{ lb}$

Steel Beam

Beam height h , Beam width b , thickness of web t_w and thickness of flange t_f .

	Item#		h	t_w	b	t_f	Area	I_{xx}	I_{yy}
		lb/ft	in	in	in	in	in ²	in ⁴	in ⁴
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3	S24x100	2674.34	24	0.745	7.245	0.87	29.3	2683.0889	55.9691
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5	S24x80	2330.69	24	0.5	7	0.87	23.5	2339.1390	49.9850
6	S20x96	1923.47	20.3	0.8	7.2	0.92	28.2	1933.7503	58.0975
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9	S20x66	1331.74	20	0.505	6.255	0.795	19.4	1337.4974	32.6410
10	S18x70	1045.64	18	0.711	6.251	0.691	20.6	1049.4203	28.6695
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12	S15x50	549.57	15	0.55	5.64	0.622	14.7	552.0609	18.8064
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14	S12x50	359.06	12	0.687	5.477	0.659	14.7	361.9356	18.3695
15	S12x40.8	315.98	12	0.462	5.252	0.659	12	318.7310	16.0100
16	S12x35	260.66	12	0.428	5.078	0.544	10.3	262.1621	11.9505
17	S12x31.8	246.37	12	0.35	5	0.544	9.4	247.8499	11.3762
18	S10x35	170.97	10	0.594	4.944	0.491	10.3	172.0457	10.0640
19	S10x25.4	140.44	10	0.311	4.661	0.491	7.46	141.4477	8.3115
20	S8x23	75.73	8	0.441	4.171	0.426	6.77	76.3200	5.2092
21	S8x18.4	66.16	8	0.271	4.001	0.426	5.41	66.7229	4.5607
22	S7x20	49.97	7	0.45	3.86	0.392	5.88	50.3990	3.8106
23	S6x17.25	31.43	6	0.465	3.565	0.359	5.07	31.7369	2.7612
24	S6x12.5	25.73	6	0.232	3.332	0.359	3.67	26.0157	2.2196
25	S5x14.75	18.55	5	0.494	3.284	0.326	4.34	18.7557	1.9745
26	S5x10	14.49	5	0.214	3.004	0.326	2.94	14.6786	1.4770
27	S4x9.5	8.30	4	0.326	2.796	0.293	2.79	8.4332	1.0790
28	S4x7.7	7.28	4	0.193	2.663	0.293	2.26	7.4054	0.9246
29	S3x7.5	3.73	3	0.349	2.509	0.26	2.21	3.8090	0.6950
30	S3x5.7	3.12	3	0.17	2.33	0.26	1.67	3.1905	0.5494

$$W = 86.54 \text{ lb} \checkmark \rightarrow$$

\$207.50 total
(10 lb/ft)

$$I_{min, steel} = \frac{(100 \text{ lbs})(72 \text{ in})^3}{3(3 \times 10^7 \text{ psi})(\frac{1}{25} \text{ in})} = 10.368 \text{ in}^4$$

$$\delta_{body} = \frac{(14.49/12 \text{ lb/in})(72 \text{ in})^4}{8(3 \times 10^7 \text{ psi})(14.6786 \text{ in}^4)} = 0.00921 \text{ in}$$

$$\delta_{max} = \frac{(100 \text{ lbs})(72 \text{ in})^3}{3(3 \times 10^7 \text{ psi})(14.6786 \text{ in}^4)} = 0.0283 \text{ in}$$

$$\delta_{body} + \delta_{max} = 0.0375 \text{ in} < \frac{1}{25} \text{ in} \checkmark$$